

NETWORK OPERATIONS STRATEGY MEMO

To: Head of Network Operations
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Our network currently operates with 93.5% of all corridor-mode-time combinations running more than 20% over estimated travel time. The analysis below identifies where the problem is concentrated, what to do about it, and what we stand to recover.

1. The Five Hubs Driving Most of Our Delays

Using network centrality analysis, we identified the five hubs that sit on the most delay-critical paths. Congestion at these nodes does not stay local; it ripples through dozens of downstream corridors. The top three alone account for 21.6% of all multi-hop traffic and 34.6% of total delay-minutes lost network-wide.

Hub	Role	Outbound routes delayed	Est. SLA breach contribution	Recommended action
IND000000ACB	National pass-through	87.2% of outbound corridors	~8-9% of network SLA breaches	Add parallel sortation path + overflow hub
IND562132AAA	Regional pass-through	94.3% of outbound corridors	~7-8% of network SLA breaches	Add parallel sortation path + overflow hub
IND501359AAE	Mid-tier pass-through	100% of outbound corridors	~6-7% of network SLA breaches	Add parallel sortation path + overflow hub
IND712311AAA	Distribution node	100% of outbound corridors	~3-4% of network SLA breaches	Monitor; likely absorbs overflow after top-3 fix
IND160002AAC	Receiving node	81.8% of outbound corridors	~2-3% of network SLA breaches	Expand inbound unloading capacity

All five hubs have roughly balanced inbound/outbound flows, meaning they are not overloaded in one direction. The fix is not about which dock to expand. It is about reducing how much traffic is forced through a single node. A secondary hub or parallel sortation path at each location is the most effective lever.

2. Corridor-Specific Interventions

Type	Scale	Intervention	Priority corridors
Facility upgrade	Top 3 hubs, 34.6% of delay-minutes	Parallel sortation + secondary overflow hub	IND000000ACB, IND562132AAA, IND501359AAE
Route-type shift (downgrade)	631 FTL corridors, delay ratio >1.5x	Move to Carting or adjust time slot	FTL lanes with >20 trips consistently overrunning
Route-type shift (upgrade)	42 Carting corridors, >70% delayed	Move to FTL for scheduling control	High-volume Carting lanes with >100 trips
Time-slot shift	Evening FTL corridors highest risk	Shift non-urgent volume to afternoon/night	ACB-to-AAA and AAA-to-ACB corridors (evening)

The three worst corridors by total delay-minutes are all FTL evening runs: IND000000ACB to IND562132AAA (31,332 delay-minutes across 2,238 trips), IND000000ACB to IND712311AAA (21,840 delay-minutes), and IND562132AAA to IND000000ACB (21,150 delay-minutes). These should be the first lanes reviewed for time-slot shifting pending facility upgrades.

3. What Upgrading the Top 3 Hubs Is Expected to Recover

Metric	Current exposure	Projected recovery (conservative)
Shipments delivered on time (within 15% of ETA)	24.7% of shipments	34.3% of shipments (+9.6 pp from model alone)
Delay-minutes attributable to top-3 hubs	647,249 minutes (34.6% of network total)	~194,175 minutes recovered (30% mitigation)
Shipments moved from late to on-time	(baseline)	~16,497 shipments per period
Revenue/penalty at risk (total network)	~INR 63.8 lakh (illustrative)	~INR 8.2 lakh recovered at top-3 hubs

These estimates use a conservative 30% mitigation assumption and a placeholder SLA penalty of INR 50 per delayed shipment. Finance should recompute using actual contract terms before budget is committed. The recovery figure does not include cascading benefits on downstream corridors, so the real number is likely higher.

Note: This decision strategy depends on historical trips' information. The secondary factors like infrastructure changes, sudden increase in seasonable volume, and availability of vehicles will not be taken into account for recommendations and the ETA model should be retrained quarterly to capture network evolution.